

Abstract No.

A transgenic glioblastoma mouse model containing a reporter element for stem-like tumor cells.

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Glioblastoma (GBM) cells endowed with stem-like properties (GSCs) can have increased potential for tumor (relapse) formation and often display accelerated resistance to standard therapy. Hence, obtaining strategies for therapeutic targeting of GSCs is a priority in neurooncology. To address this issue, we developed a transgenic mouse model faithfully recapitulating human disease and also containing a reporter function for GSCs. Histopathologically, the experimental tumors contain all hallmarks of human GBM. Tumors are induced after subventricular injection of a hybrid AAV (providing flexibility for the time-point of tumor initiation) and the vector assures target-specific recombination (in neural precursor cells). The vector contains sufficient payload in order to accommodate different proto-oncogenes or can drive the expression of imaging markers like luciferase in order to monitor tumor growth in real-time. Furthermore, cre-recombinase induced expression of RFP allows tracing of tumor cells on the clonal level. Cross-breeding the model with another transgenic mouse strain harboring a modified nestin-promoter element driving GFP expression resulted in combined RFP/GFP expression in a subpopulation of GBM cells. Purification of RFP+/GFP+ cells and orthotopic re-implantation of sorted cells in three consecutive experimental cycles revealed that 500 RFP+/GFP+ are sufficient to generate GBM whereas inoculation of 50,000 RFP+/GFP-negative GBM cells was insufficient to induce tumorigenesis. In addition to their self-renewal capacity throughout in vivo experimentation, the RFP+/GFP+ also showed accelerated self-renewal in vitro (in a limiting dilution assay). Cultivation of RFP+/GFP+ cells over several passages and recapitulation of the serial implantation experiment indicated that RFP+/GFP+ retain their stem-like capacity throughout in vitro maintenance. Overall, we describe an experimental model with profound potential for translational research on GBM.



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